

Genome-wide fusion-hotspot cohort scan: msk_impact_50k_2026

544 of 3919 genes passed the ≥ 5 -patient recurrence gate; 1 FDR-significant ($q < 0.05$).

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Gene	Source	Status	Patients	In-frame%	Domain-ret. %	Best q	Sig.	Composite
ETV6	auto	ok	213	71.11	75.56	0.004334	yes	0.4158
RET	curated	ok	219	75.26	92.27	0.1198	no	0.4372
FGFR2	auto	ok	188	79.41	84.56	0.1198	no	0.3603
ERG	auto	ok	863	30.2	96.07	0.1821	no	0.4274
EWSR1	auto	ok	415	65.33	11.17	0.1821	no	0.2229
ALK	curated	ok	321	83.09	95.96	0.1821	no	0.6098
TP53	auto	ok	258	8.511	25.53	0.1821	no	0.2508
EML4	auto	ok	224	87.89	53.36	0.1821	no	0.6283
ROS1	auto	ok	204	59.02	87.7	0.1821	no	0.4618
FGFR3	auto	ok	194	48.68	93.42	0.1821	no	0.6233
TACC3	auto	ok	138	50	48.48	0.1821	no	0.5258
NOTCH1	auto	ok	115	23.4	59.57	0.1821	no	0.3796
NTRK1	curated	ok	113	41.03	82.05	0.1821	no	0.3201
NAB2	auto	ok	111	34.72	68.06	0.1821	no	0.4606
CDK12	auto	ok	99	16.28	37.21	0.1821	no	0.252
TERT	auto	ok	92	0	68.42	0.1821	no	0.3993
CD74	auto	ok	74	58.21	88.06	0.1821	no	0.4234
MET	auto	ok	73	51.52	72.73	0.1821	no	0.3248
NTRK3	auto	ok	71	94.83	94.83	0.1821	no	0.478
STAT6	auto	ok	60	43.4	5.66	0.1821	no	0.4286
DNAJB1	auto	ok	54	64.1	2.564	0.1821	no	0.6424
RNF43	auto	ok	48	28.57	33.33	0.1821	no	0.2347
KIAA1549	auto	ok	44	90.91	50	0.1821	no	0.4904
CDKN2B	auto	ok	40	16.67	16.67	0.1821	no	0.4303
POLD1	auto	ok	36	12.5	37.5	0.1821	no	0.2807
RBM10	auto	ok	36	42.86	71.43	0.1821	no	0.3471
PRKACA	auto	ok	35	68.57	82.86	0.1821	no	0.7022
ATF1	auto	ok	30	70	90	0.1821	no	0.4859
BICC1	auto	ok	29	96.43	67.86	0.1821	no	0.4286
RPS6KA4	auto	ok	21	38.46	53.85	0.1821	no	0.2556

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
LMNA	auto	ok	17	46.15	38.46	0.1821	no	0.4387
AGK	auto	ok	14	92.86	14.29	0.1821	no	0.6366
TPM3	auto	ok	14	78.57	64.29	0.1821	no	0.5289
EMID1	auto	ok	5	75	25	0.1821	no	0.5002
EGFR	auto	ok	466	50.91	74.55	0.2062	no	0.3716
BRAF	curated	ok	247	84.36	91.06	0.2356	no	0.2273
TSC2	auto	ok	88	17.65	23.53	0.2366	no	0.2352
PRKD1	auto	ok	16	50	25	0.2366	no	0.3877
TMPRSS2	auto	ok	1088	29.99	6.113	0.2771	no	0.5539
FLI1	auto	ok	121	61.86	5.932	0.2771	no	0.58
NCOR1	auto	ok	80	0	31.58	0.2771	no	0.3684
AGO2	auto	ok	49	26.09	30.43	0.2771	no	0.3988
KDR	auto	ok	28	11.11	33.33	0.2771	no	0.2619
SLX4	auto	ok	28	26.32	57.89	0.2771	no	0.3937
IKBKE	auto	ok	27	25	33.33	0.2771	no	0.3108
CASP8	auto	ok	19	22.22	55.56	0.2771	no	0.3226
NOTCH2	auto	ok	92	3.571	50	0.3155	no	0.2735
CCDC6	auto	ok	53	92.45	1.887	0.3155	no	0.4124
SUFU	auto	ok	17	25	25	0.3329	no	0.2691
BCAS3	auto	ok	18	6.25	43.75	0.3532	no	0.3214
PAX8	auto	ok	43	30	80	0.392	no	0.2831
INPPL1	auto	ok	39	28.57	50	0.4047	no	0.2793
ACOXL	auto	ok	8	12.5	50	0.4226	no	0.389
TOP1	auto	ok	31	0	50	0.462	no	0.4115
FGFR1	auto	ok	64	30.43	47.83	0.4754	no	0.2332
MAP3K1	auto	ok	42	10	30	0.4754	no	0.2639
CARM1	auto	ok	39	22.22	27.78	0.4754	no	0.2547
PRKAR1A	auto	ok	12	33.33	16.67	0.4754	no	0.3342
CREBBP	auto	ok	121	13.04	32.61	0.4971	no	0.2884
ERBB3	auto	ok	45	11.11	55.56	0.4971	no	0.2632
TSC1	auto	ok	28	0	57.14	0.5246	no	0.4286
NRG1	auto	ok	16	53.33	86.67	0.5474	no	0.2839
FLT4	auto	ok	35	14.29	42.86	0.5603	no	0.2861
ABL1	auto	ok	21	0	22.22	0.613	no	0.4074
FH	auto	ok	29	6.25	12.5	0.6259	no	0.2604
ELF3	auto	ok	47	27.27	45.45	0.6339	no	0.3275
RAD51B	auto	ok	16	50	50	0.6339	no	0.3247

Gene	Source	Status	Patients	In-frame%	Domain-ret.%	Best q	Sig.	Composite
AKT2	auto	ok	22	10	20	0.6448	no	0.2571
ARID1B	auto	ok	72	0	53.85	0.6468	no	0.5385
GOPC	auto	ok	6	75	50	0.6488	no	0.4686
PLCG2	auto	ok	17	33.33	66.67	0.7099	no	0.3241
EXOC4	auto	ok	5	0	40	0.7113	no	0.6
STAT5B	auto	ok	27	15.38	69.23	0.7954	no	0.2803
ESR1	auto	ok	30	23.08	38.46	0.8453	no	0.282
PIK3CB	auto	ok	20	0	60	0.8453	no	0.5021
SND1	auto	ok	17	94.12	17.65	0.8453	no	0.4286
CD79A	auto	ok	13	0	37.5	0.8453	no	0.4167
SDHA	auto	ok	12	40	60	0.8453	no	0.3344
USP34	auto	ok	7	14.29	71.43	0.8453	no	0.4286
NF1	auto	ok	247	10.64	44.68	0.8758	no	0.2564
CIC	auto	ok	68	0	42.86	0.8758	no	0.3651
KDM5A	auto	ok	44	30	50	0.8758	no	0.2794
ATM	auto	ok	72	3.704	29.63	0.8999	no	0.2343
BRCA1	auto	ok	66	7.407	48.15	0.8999	no	0.2348
TFE3	auto	ok	45	56.76	78.38	0.8999	no	0.413
PPM1D	auto	ok	27	8.333	25	0.8999	no	0.25
NELL1	auto	ok	5	20	80	0.8999	no	0.4176
RB1	auto	ok	147	0	47.37	0.9234	no	0.4011
KMT2A	auto	ok	65	19.05	47.62	0.9234	no	0.2381
RAD21	auto	ok	38	44.44	55.56	0.9234	no	0.341
PICALM	auto	ok	5	0	60	0.9234	no	0.4667
NF2	auto	ok	46	8.333	8.333	0.9665	no	0.2566
NOTCH4	auto	ok	44	0	68.75	0.9665	no	0.375
RAF1	auto	ok	23	25	50	0.9665	no	0.25
SMARCA4	auto	ok	143	19.44	41.67	0.9885	no	0.2958
KMT2D	auto	ok	128	2.128	36.17	0.9885	no	0.2325
WT1	auto	ok	119	89.42	4.808	0.9885	no	0.6202
EP300	auto	ok	98	4.545	31.82	0.9885	no	0.2394
STK11	auto	ok	97	4.348	30.43	0.9885	no	0.2498
ERBB2	auto	ok	88	15.91	43.18	0.9885	no	0.2183
POLE	auto	ok	73	18.18	31.82	0.9885	no	0.2315
MGA	auto	ok	65	6.667	46.67	0.9885	no	0.2429
STAT3	auto	ok	51	8.333	8.333	0.9885	no	0.2518
RICTOR	auto	ok	34	30	50	0.9885	no	0.2858

Gene	Source	Status	Patients	In-frame%	Domain-ret. %	Best q	Sig.	Composite
DNMT3A	auto	ok	32	36.36	36.36	0.9885	no	0.3395
INSR	auto	ok	31	16.67	50	0.9885	no	0.2877
XPO1	auto	ok	31	8.333	41.67	0.9885	no	0.2572
FUBP1	auto	ok	29	18.18	54.55	0.9885	no	0.2592
RARA	auto	ok	28	28.57	35.71	0.9885	no	0.2449
PAK1	auto	ok	25	10	10	0.9885	no	0.2571
SMARCB1	auto	ok	23	0	60	0.9885	no	0.5993
BLM	auto	ok	20	0	50	0.9885	no	0.4444
GRIN2A	auto	ok	19	0	16.67	0.9885	no	0.4444
SEC16A	auto	ok	17	23.53	41.18	0.9885	no	0.4286
TTC28	auto	ok	15	16.67	58.33	0.9885	no	0.2583
NUF2	auto	ok	14	40	10	0.9885	no	0.2571
FBXL20	auto	ok	8	28.57	28.57	0.9885	no	0.381
MTAP	auto	ok	8	25	25	0.9885	no	0.4601
KIF5B	auto	ok	95	88.54	96.88	0.9895	no	0.5487
RTEL1	auto	ok	57	20.83	37.5	0.9895	no	0.2573
AXIN1	auto	ok	29	50	66.67	0.9895	no	0.4238
PRKCI	auto	ok	22	33.33	50	0.9895	no	0.3719
GAB2	auto	ok	7	20	20	0.9895	no	0.3
ZNRF3	auto	ok	6	33.33	66.67	0.9895	no	0.4618
ARID1A	auto	ok	148	12.9	35.48	1	no	0.2354
KMT2C	auto	ok	134	6.25	46.88	1	no	0.2359
APC	auto	ok	110	0	45.45	1	no	0.4992
NOTCH3	auto	ok	101	3.846	50	1	no	0.2308
FAT1	auto	ok	95	0	27.27	1	no	0.4545
BRD4	auto	ok	86	3.125	34.38	1	no	0.1991
PTEN	auto	ok	86	22.22	22.22	1	no	0.2619
DOT1L	auto	ok	85	22.73	36.36	1	no	0.2356
PBRM1	auto	ok	85	5.556	50	1	no	0.2533
ZFXH3	auto	ok	85	0	14.29	1	no	0.5238
DNMT1	auto	ok	80	17.86	17.86	1	no	0.263
BRCA2	auto	ok	79	8.333	58.33	1	no	0.25
ARID2	auto	ok	72	0	23.08	1	no	0.3946
CTNNB1	auto	ok	72	30	40	1	no	0.2604
KDM6A	auto	ok	71	25	75	1	no	0.06818
BAP1	auto	ok	67	16.67	33.33	1	no	0.2381
ATRX	auto	ok	64	28.57	42.86	1	no	0.2871

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SMAD4	auto	ok	62	0	16.67	1	no	0.4444
SPEN	auto	ok	61	0	26.32	1	no	0.357
CDH1	auto	ok	58	11.11	66.67	1	no	0.2627
PTPRD	auto	ok	57	12.5	37.5	1	no	0.2715
PIK3R1	auto	ok	55	33.33	33.33	1	no	0.1818
ANKRD11	auto	ok	54	0	66.67	1	no	0.4444
MDC1	auto	ok	52	7.143	57.14	1	no	0.2501
FOXA1	auto	ok	51	0	18.18	1	no	0.5969
GLI1	auto	ok	51	19.05	57.14	1	no	0.2395
FANCA	auto	ok	49	15	35	1	no	0.2201
MTOR	auto	ok	49	10	30	1	no	0.2334
ATR	auto	ok	48	0	28.57	1	no	0.4286
BRIP1	auto	ok	46	0	21.05	1	no	0.3935
SETD2	auto	ok	46	0	47.06	1	no	0.3725
NSD1	auto	ok	45	8.333	33.33	1	no	0.3259
PTPRS	auto	ok	44	7.143	35.71	1	no	0.2449
PIK3C2G	auto	ok	43	30	30	1	no	0.2961
RPTOR	auto	ok	43	23.08	15.38	1	no	0.2637
B2M	auto	ok	42	0	0	1	no	0.7386
ERBB4	auto	ok	42	40	40	1	no	0.3008
DNMT3B	auto	ok	40	0	14.29	1	no	0.4286
KEAP1	auto	ok	40	0	25	1	no	0.4167
NFE2L2	auto	ok	40	0	20	1	no	0.5653
EZH2	auto	ok	39	14.29	7.143	1	no	0.3478
FOXP1	auto	ok	39	20	80	1	no	0.3
NBN	auto	ok	38	0	33.33	1	no	0.4444
AXL	auto	ok	37	30	20	1	no	0.2657
PREX2	auto	ok	37	0	50	1	no	0.5
BCOR	auto	ok	36	0	57.14	1	no	0.4286
DROSHA	auto	ok	36	42.86	14.29	1	no	0.3117
IGF1R	auto	ok	36	8.333	25	1	no	0.25
TBX3	auto	ok	36	0	50	1	no	0.4444
KDM5C	auto	ok	33	30	60	1	no	0.2586
NCOA3	auto	ok	33	23.08	69.23	1	no	0.2545
PTPRT	auto	ok	33	33.33	50	1	no	0.2862
RUNX1	auto	ok	33	16.67	33.33	1	no	0.2865
BCL2L11	auto	ok	32	21.43	21.43	1	no	0.3171

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JAK3	auto	ok	32	0	33.33	1	no	0.4074
PIK3R2	auto	ok	32	10	40	1	no	0.2571
RPS6KB2	auto	ok	32	33.33	0	1	no	0.2619
ARID5B	auto	ok	31	0	33.33	1	no	0.5019
CSDE1	auto	ok	31	0	60	1	no	0.4667
ERCC2	auto	ok	31	25	75	1	no	0.269
PDGFRA	auto	ok	31	55.56	55.56	1	no	0.2657
TP53BP1	auto	ok	31	18.18	54.55	1	no	0.285
ETV1	auto	ok	30	52.38	0	1	no	0.3636
EZH1	auto	ok	30	0	18.18	1	no	0.3939
GNAS	auto	ok	30	25	0	1	no	0.2679
CALR	auto	ok	29	44.44	0	1	no	0.2619
CBL	auto	ok	29	25	50	1	no	0.3614
JAK1	auto	ok	29	12.5	0	1	no	0.2679
NTRK2	auto	ok	29	0	40	1	no	0.4667
AXIN2	auto	ok	28	0	75	1	no	0.5
TEK	auto	ok	28	0	55.56	1	no	0.4074
TET1	auto	ok	28	14.29	28.57	1	no	0.2755
MAP2K4	auto	ok	27	14.29	0	1	no	0.3281
SMAD3	auto	ok	27	11.11	55.56	1	no	0.3042
TRAF7	auto	ok	27	10	20	1	no	0.3132
IRS2	auto	ok	26	11.11	77.78	1	no	0.2624
SOX9	auto	ok	26	0	20	1	no	0.4667
UPF1	auto	ok	26	33.33	50	1	no	0.3017
CARD11	auto	ok	25	0	28.57	1	no	0.4286
FLT1	auto	ok	25	33.33	33.33	1	no	0.1957
MALT1	auto	ok	25	42.86	42.86	1	no	0.2755
ASXL2	auto	ok	24	28.57	57.14	1	no	0.3476
DIS3	auto	ok	24	16.67	16.67	1	no	0.4286
TP63	auto	ok	24	28.57	14.29	1	no	0.2755
AKT1	auto	ok	23	0	42.86	1	no	0.5238
MAP3K13	auto	ok	23	0	20	1	no	0.475
MDM2	auto	ok	23	23.53	47.06	1	no	0.2447
MED12	auto	ok	23	0	33.33	1	no	0.4637
STAT5A	auto	ok	23	7.692	0	1	no	0.2473
TCF7L2	auto	ok	23	0	50	1	no	0.5
EIF4A2	auto	ok	22	0	33.33	1	no	0.4444

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PARP1	auto	ok	22	10	40	1	no	0.3174
PIK3CD	auto	ok	22	12.5	50	1	no	0.2693
PLK2	auto	ok	22	28.57	57.14	1	no	0.3391
PPP2R1A	auto	ok	22	50	33.33	1	no	0.299
RAD54L	auto	ok	22	0	60	1	no	0.4667
FLT3	auto	ok	21	0	20	1	no	0.4667
MAP2K2	auto	ok	21	28.57	14.29	1	no	0.23
SMARCD1	auto	ok	21	12.5	50	1	no	0.2771
ASPSCR1	auto	ok	20	70	20	1	no	0.4071
FLCN	auto	ok	20	50	40	1	no	0.2626
MSH6	auto	ok	20	0	66.67	1	no	0.4444
PIK3R3	auto	ok	20	33.33	0	1	no	0.1818
SMYD3	auto	ok	20	25	25	1	no	0.1364
SRC	auto	ok	20	25	0	1	no	0.3214
CHEK2	auto	ok	19	16.67	16.67	1	no	0.3488
CSF1R	auto	ok	19	0	33.33	1	no	0.4074
LATS1	auto	ok	19	16.67	33.33	1	no	0.2894
PTCH1	auto	ok	19	0	20	1	no	0.4667
YAP1	auto	ok	19	20	20	1	no	0.3
FGFR4	auto	ok	18	11.11	44.44	1	no	0.2619
PGR	auto	ok	18	14.29	57.14	1	no	0.2763
VTCN1	auto	ok	18	20	60	1	no	0.3021
CSF3R	auto	ok	17	33.33	66.67	1	no	0.2973
EGFL7	auto	ok	17	50	25	1	no	0.4286
ERCC5	auto	ok	17	0	37.5	1	no	0.5
LDLR	auto	ok	17	20	53.33	1	no	0.4829
SDC4	auto	ok	17	82.35	0	1	no	0.4286
TAP1	auto	ok	17	0	42.86	1	no	0.4286
TGFBR1	auto	ok	17	25	75	1	no	0.3583
CTCF	auto	ok	16	0	50	1	no	0.5
CUL1	auto	ok	16	35.71	0	1	no	0.3506
PDGFRB	auto	ok	16	40	40	1	no	0.3378
RECQL	auto	ok	16	0	0	1	no	0.5
SMO	auto	ok	16	0	12.5	1	no	0.4167
SUZ12	auto	ok	16	0	20	1	no	0.4667
CCND1	auto	ok	15	0	57.14	1	no	0.4286
CCNE1	auto	ok	15	20	0	1	no	0.3

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MSH3	auto	ok	15	0	50	1	no	0.5
TNFRSF14	auto	ok	15	25	0	1	no	0.2727
CDK4	auto	ok	14	33.33	0	1	no	0.09091
E2F3	auto	ok	14	0	20	1	no	0.4667
FOXO1	auto	ok	14	50	0	1	no	0.3763
MPL	auto	ok	14	0	83.33	1	no	0.4444
NUP93	auto	ok	14	0	20	1	no	0.4667
SESN3	auto	ok	14	20	0	1	no	0.1091
SF3B1	auto	ok	14	20	40	1	no	0.3
EZR	auto	ok	13	61.54	0	1	no	0.5455
MST1R	auto	ok	13	20	40	1	no	0.3122
NPM1	auto	ok	13	50	50	1	no	0.333
TRAPPC9	auto	ok	13	23.08	0	1	no	0.5035
YES1	auto	ok	13	16.67	16.67	1	no	0.314
AKT3	auto	ok	12	33.33	33.33	1	no	0.3636
CEBPA	auto	ok	12	0	0	1	no	0.4444
DUSP4	auto	ok	12	0	33.33	1	no	0.4444
FYN	auto	ok	12	66.67	66.67	1	no	0.1822
IDH1	auto	ok	12	25	0	1	no	0.2727
PAX5	auto	ok	12	33.33	66.67	1	no	0.1822
PIK3C3	auto	ok	12	100	100	1	no	0.5455
PIK3CA	auto	ok	12	0	80	1	no	0.5924
ERCC4	auto	ok	11	20	40	1	no	0.3
FGF19	auto	ok	11	0	0	1	no	0.5
GSK3B	auto	ok	11	50	0	1	no	0.2727
MKRN1	auto	ok	11	63.64	0	1	no	0.4959
MLH1	auto	ok	11	50	50	1	no	0.2727
PHF7	auto	ok	11	50	0	1	no	0.2727
RAD51D	auto	ok	11	50	0	1	no	0.2727
RELA	auto	ok	11	33.33	100	1	no	0.4545
SOS1	auto	ok	11	33.33	0	1	no	0.1818
TACC2	auto	ok	11	62.5	100	1	no	0.5455
BMPR1A	auto	ok	10	100	0	1	no	0.5455
CDK8	auto	ok	10	25	0	1	no	0.1364
CUL3	auto	ok	10	50	0	1	no	0.2727
EPAS1	auto	ok	10	0	20	1	no	0.4667
IKZF3	auto	ok	10	25	62.5	1	no	0.2679

Gene	Source	Status	Patients	In-frame%	Domain-ret. %	Best q	Sig.	Composite
MDM4	auto	ok	10	25	50	1	no	0.329
RXRA	auto	ok	10	33.33	66.67	1	no	0.1818
TRIM24	auto	ok	10	88.89	0	1	no	0.4848
ARAF	auto	ok	9	50	50	1	no	0.2727
LRP1	auto	ok	9	0	20	1	no	0.4854
NTHL1	auto	ok	9	33.33	66.67	1	no	0.1942
PPARG	auto	ok	9	50	50	1	no	0.2827
RAD51	auto	ok	9	66.67	66.67	1	no	0.3664
SESN1	auto	ok	9	0	0	1	no	0.4667
STK40	auto	ok	9	33.33	0	1	no	0.1818
ETV4	auto	ok	8	33.33	16.67	1	no	0.396
HNF1A	auto	ok	8	0	20	1	no	0.4667
IKZF1	auto	ok	8	100	0	1	no	0.5455
IRF4	auto	ok	8	0	20	1	no	0.4667
KRAS	auto	ok	8	33.33	0	1	no	0.1818
MAPK1	auto	ok	8	50	0	1	no	0.2727
MCL1	auto	ok	8	0	50	1	no	0.4538
MSI1	auto	ok	8	20	40	1	no	0.2689
MYD88	auto	ok	8	33.33	33.33	1	no	0.2933
PDE4D	auto	ok	8	16.67	66.67	1	no	0.3801
PTK2	auto	ok	8	25	50	1	no	0.3447
SLC25A21	auto	ok	8	0	12.5	1	no	0.5
STRN	auto	ok	8	25	0	1	no	0.5661
CACNA1A	auto	ok	7	14.29	42.86	1	no	0.2844
CDK6	auto	ok	7	50	0	1	no	0.2727
CREB1	auto	ok	7	57.14	71.43	1	no	0.4446
CRKL	auto	ok	7	0	50	1	no	0.6441
DCUN1D1	auto	ok	7	66.67	0	1	no	0.1818
DGKH	auto	ok	7	0	28.57	1	no	0.4762
DNM2	auto	ok	7	28.57	42.86	1	no	0.2923
IGF2	auto	ok	7	50	75	1	no	0.3245
KREMEN1	auto	ok	7	28.57	42.86	1	no	0.2755
NR4A3	auto	ok	7	57.14	71.43	1	no	0.4286
ST7	auto	ok	7	14.29	0	1	no	0.4675
TACC1	auto	ok	7	25	75	1	no	0.2727
CAMTA1	auto	ok	6	20	0	1	no	0.1091
CDKN2C	auto	ok	6	33.33	66.67	1	no	0.2039

Gene	Source	Status	Patients	In-frame%	Domain-ret. %	Best q	Sig.	Composite
CPM	auto	ok	6	40	0	1	no	0.1091
CTNNA3	auto	ok	6	60	0	1	no	0.3273
DTNB	auto	ok	6	16.67	0	1	no	0.3571
EIF3H	auto	ok	6	60	0	1	no	0.5455
EPHX3	auto	ok	6	16.67	0	1	no	0.2727
NOS1AP	auto	ok	6	0	16.67	1	no	0.4444
SMARCA2	auto	ok	6	33.33	66.67	1	no	0.1857
SOX2	auto	ok	6	0	75	1	no	0.5009
TANC2	auto	ok	6	16.67	50	1	no	0.2857
TFG	auto	ok	6	100	100	1	no	0.2727
ATE1	auto	ok	5	33.33	0	1	no	0.5455
BCL2L1	auto	ok	5	50	50	1	no	0.2866
CNTNAP2	auto	ok	5	66.67	0	1	no	0.3636
CREM	auto	ok	5	20	100	1	no	0.3273
CTTNBP2	auto	ok	5	0	20	1	no	0.6
EIF4E	auto	ok	5	50	0	1	no	0.2727
FBXL7	auto	ok	5	50	75	1	no	0.4366
GPATCH8	auto	ok	5	25	50	1	no	0.3239
HCN1	auto	ok	5	100	0	1	no	0.5455
LRP1B	auto	ok	5	50	50	1	no	0.2727
MAD1L1	auto	ok	5	40	0	1	no	0.2182
PRKAG2	auto	ok	5	20	40	1	no	0.3429
RAC2	auto	ok	5	50	0	1	no	0.2727
SBNO2	auto	ok	5	0	75	1	no	0.5
SDHAF2	auto	ok	5	100	0	1	no	0.5455
SFPQ	auto	ok	5	25	75	1	no	0.4286
SHANK2	auto	ok	5	0	25	1	no	0.5
SMARCE1	auto	ok	5	100	0	1	no	0.5455
SMURF2	auto	ok	5	25	75	1	no	0.3374
SPTBN1	auto	ok	5	50	0	1	no	0.5455
THSD4	auto	ok	5	25	25	1	no	0.3214
TPR	auto	ok	5	40	0	1	no	0.5455
WWOX	auto	ok	5	0	40	1	no	0.4667
CDKN2A	auto	ok	185	4.762	0		no	0.2857
KMT2B	unresolved	failed	68				no	
NSD3	auto	ok	58	14.29	0		no	0.1429
RECQL4	unresolved	failed	54				no	

Gene	Source	Status	Patients	In-frame%	Domain-ret. %	Best q	Sig.	Composite
PRKN	auto	ok	40	50	0		no	0.25
NSD2	auto	ok	37	33.33	0		no	0.3333
MAP3K14	unresolved	failed	36				no	
ASXL1	auto	ok	35	0	77.78		no	0.2222
TCF3	unresolved	failed	35				no	
RASA1	auto	ok	32	0	50		no	0.5
STAG2	auto	ok	31	0	0		no	0.5
AR	auto	ok	30	0	50		no	0.5
CDC73	auto	ok	28	0	50		no	0.5
JAK2	auto	ok	27	0	33.33		no	0.3333
CDKN1B	auto	ok	24	0	0		no	0.3333
MSH2	auto	ok	23	0	0		no	0.25
NFKBIA	auto	ok	23	0	100		no	1
MEN1	auto	ok	22	0	0		no	0.3333
MYC	auto	ok	22	0	0		no	0.3333
TET2	auto	ok	21	0	66.67		no	0.6667
ERF	unresolved	failed	20				no	
KIT	auto	ok	20	0	0		no	0.5
CDKN2B-AS1	unresolved	failed	19				no	
EPHA3	auto	ok	19	0	0		no	0.5
NEGR1	auto	ok	19	0	66.67		no	0.3333
PALB2	auto	ok	19	0	0		no	0.2
SEPTIN14	unresolved	failed	19				no	
BBC3	auto	ok	18	50	0		no	0.375
GATA3	auto	ok	18	0	0		no	0.5
AMER1	auto	ok	17	0	33.33		no	0.3333
COP1	auto	ok	17	50	0		no	0.5
DDR2	auto	ok	17	0	0		no	0.2
EPHA5	auto	ok	17	0	33.33		no	0.6667
INPP4B	auto	ok	17	100	0		no	1
RAD50	auto	ok	17	0	33.33		no	0.3333
BCL6	auto	ok	16	0	66.67		no	0.3333
DICER1	auto	ok	16	0	0		no	0.4
ERRFI1	auto	ok	16	0	0		no	0.6667
LATS2	auto	ok	16	0	0		no	1
MITF	auto	ok	16	0	100		no	1
MSI2	auto	ok	16	0	50		no	0.5

Gene	Source	Status	Patients	In-frame%	Domain-ret. %	Best q	Sig.	Composite
NCOA4	auto	failed	16				no	
RAD52	auto	ok	16	0	100		no	1
TGFBR2	auto	ok	16	0	50		no	0.5
TRAP1	unresolved	failed	16				no	
CREB3L1	unresolved	failed	15				no	
INPP4A	auto	ok	15	14.29	0		no	0.2857
SRP19	auto	ok	15	0	0		no	1
GPS2	auto	ok	14	0	0		no	0.5
HGF	auto	ok	14	0	50		no	0.5
LINC00114	unresolved	failed	14				no	
RIT1	auto	ok	14	0	0		no	0.3333
TAP2	auto	ok	14	0	0		no	0.5
TRAF2	unresolved	failed	14				no	
CYLD	auto	ok	13	0	0		no	1
DAXX	auto	ok	13	0	0		no	1
MAP2K1	auto	ok	13	0	50		no	0.25
MEF2B	unresolved	failed	13				no	
MRE11	auto	ok	13	0	20		no	0.2
PAK5	auto	ok	13	50	0		no	0.5
RAD51C	auto	ok	13	0	0		no	0.1667
SPOP	auto	ok	13	0	50		no	0.5
BIRC3	auto	ok	12	0	66.67		no	0.3333
BTK	auto	ok	12	0	0		no	1
CBFB	auto	ok	12	0	0		no	1
EED	auto	ok	12	0	0		no	0.5
IRS1	auto	ok	12	0	0		no	0.5
RAB35	auto	ok	12	0	0		no	0.3333
SMAD2	auto	ok	12	0	0		no	1
VEGFA	auto	ok	12	0	0		no	0.5
ALOX12B	auto	ok	11	0	0		no	0.5
BABAM1	auto	ok	11	0	0		no	0.5
BCL2	auto	ok	11	0	0		no	1
EPHA7	auto	ok	11	0	50		no	0.5
FANCC	auto	ok	11	0	0		no	0.6667
FBXW7	auto	ok	11	0	0		no	0.3333
IDH2	auto	ok	11	0	0		no	0.5
SRSF2	auto	ok	11	0	100		no	1

Gene	Source	Status	Patients	In-frame%	Domain-ret. %	Best q	Sig.	Composite
TNFAIP3	auto	ok	11	0	0		no	0.3333
AURKA	auto	ok	10	0	33.33		no	0.6667
CCND2	auto	ok	10	0	60		no	0.4
EPHB1	auto	ok	10	0	50		no	0.5
H3C2	unresolved	failed	10				no	
MAPK3	auto	ok	10	0	0		no	1
PIK3CG	auto	ok	10	0	0		no	
PMS1	auto	ok	10	0	0		no	
PTPN11	auto	ok	10	0	0		no	0.3333
SH2B3	auto	ok	10	0	25		no	0.25
SOX17	auto	ok	10	0	0		no	0.6667
SYK	auto	ok	10	0	0		no	1
CHEK1	auto	ok	9	0	0		no	0.5
GATA1	auto	ok	9	0	0		no	1
IFNGR1	auto	ok	9	0	33.33		no	0.3333
MAPKAP1	auto	ok	9	0	100		no	1
PPP6C	auto	ok	9	0	0		no	1
PRDM14	auto	ok	9	0	0		no	
REL	auto	ok	9	0	0		no	
SHQ1	auto	ok	9	0	50		no	0.5
AKAP8	auto	ok	8	0	0		no	1
AURKB	auto	ok	8	0	33.33		no	0.3333
BARD1	auto	ok	8	0	0		no	
BCL2L14	auto	ok	8	50	0		no	0.75
CCND3	auto	ok	8	0	0		no	1
CD79B	auto	ok	8	0	33.33		no	0.3333
ERCC3	auto	ok	8	0	33.33		no	0.3333
FGF3	auto	ok	8	0	0		no	0.3333
GNA11	auto	ok	8	0	0		no	1
ICOSLG	auto	ok	8	0	0		no	
LYN	auto	ok	8	0	0		no	0.25
NBPF20	unresolved	failed	8				no	
PDCD1	auto	ok	8	0	0		no	1
RRAS	auto	ok	8	0	33.33		no	0.3333
XRCC2	auto	ok	8	0	0		no	1
ACP3	auto	ok	7	71.43	0		no	0.4286
AP1B1	auto	ok	7	0	0		no	1

Gene	Source	Status	Patients	In-frame%	Domain-ret. %	Best q	Sig.	Composite
ASIC2	auto	ok	7	16.67	0		no	0.3333
ATP1B2	auto	ok	7	0	0		no	1
DCBLD1	auto	ok	7	0	0		no	
EPCAM	auto	ok	7	0	0		no	0.5
H3C4	auto	ok	7	0	0		no	
INHA	auto	ok	7	0	0		no	1
JUN	auto	ok	7	0	0		no	
KLF4	auto	ok	7	0	0		no	
NKX3-1	auto	ok	7	0	0		no	1
PDPK1	auto	ok	7	0	100		no	1
RAB11FIP4	unresolved	failed	7				no	
RYBP	auto	ok	7	0	0		no	
SDHC	auto	ok	7	0	66.67		no	0.3333
SHOC2	auto	ok	7	0	100		no	1
TENT5C	auto	ok	7	0	0		no	
AGAP3	unresolved	failed	6				no	
CD274	auto	ok	6	0	0		no	1
CD276	auto	ok	6	0	0		no	
CDH3	unresolved	failed	6				no	
FGF4	auto	ok	6	0	50		no	0.5
GALNT11	auto	ok	6	0	33.33		no	1
GRB7	auto	ok	6	0	20		no	0.8
H1-2	auto	ok	6	0	0		no	1
H3-3B	auto	ok	6	0	0		no	0.5
H3C6	unresolved	failed	6				no	
HOXB13	auto	ok	6	0	0		no	0.5

Showing the first 500 of 544 scanned genes.

Honorable mentions: highly ranked non-FDR-significant genes worth human review

Did NOT survive genome-wide FDR correction -- ranked by raw Fisher p-value among non-significant genes. Not a claim of significance.

Rank	Gene	Fisher p	Best q	N events	In-frame%	Domain-ret. %
1	RET	0.0004197	0.1198	194	75.26	92.27
2	FGFR2	0.0004247	0.1198	136	79.41	84.56

Rank	Gene	Fisher p	Best q	N events	In-frame%	Domain-ret.%
3	ALK	0.001917	0.1821	272	83.09	95.96
4	EGFR	0.01145	0.2062	55	50.91	74.55
5	BRAF	0.01337	0.2356	179	84.36	91.06
6	FGFR3	0.01948	0.1821	152	48.68	93.42
7	CDKN2B	0.02222	0.1821	12	16.67	16.67
8	IKBKE	0.02424	0.2771	12	25	33.33
9	TMPRSS2	0.0292	0.2771	867	29.99	6.113
10	NTRK1	0.04826	0.1821	78	41.03	82.05
11	INPPL1	0.04895	0.4047	14	28.57	50
12	CDK12	0.0677	0.1821	43	16.28	37.21
13	NTRK3	0.06955	0.1821	58	94.83	94.83
14	NAB2	0.1126	0.1821	72	34.72	68.06
15	KDM5A	0.119	0.8758	10	30	50

Figures

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